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Department of Computer Science and Business Systems

Academic Year 2024–2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech CSBS
Course Code & Title: CS3492 Database Management Systems
Name of the Faculty member: Dr.S.Erana Veerappa Dinesh, ASP/CSBS

Innovative Practice Description

- **Unit/Topic:** Unit 4/ B+ tree Index Files
- **Course Outcome:** CO4
- **Activity Chosen:** Jigsaw
- **Justification:**
  The concept of B+ Tree Index Files is a vital topic in database indexing, used to enhance the efficiency of data retrieval operations. It plays a crucial role in query optimization by providing a balanced tree structure that supports dynamic insertion, deletion, and search operations in logarithmic time. Understanding how B+ Trees work is essential for students to grasp how modern databases manage and access large volumes of data efficiently. Through collaborative learning, students can work together to implement B+ Tree operations and simulate indexing scenarios, which deepens their understanding of abstract data structures and real-world database performance. This approach not only strengthens their technical skills but also boosts their teamwork, problem-solving, and communication abilities. It makes learning more interactive and engaging, setting the topic apart from the traditional lecture-based approach.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
  - Allowed the students to make the group on own.
  - Individual problem statements were given to the students prior to the start of the event for preparation purposes.
  - All the students actively shared their views on the allotted topics.
  - Finally, faculty member consolidated the information that was discussed in this activity.

CO–PO/ PSO mapping:

CO	PO1	PO2	PO3	PO10	PSO1
CO1	3	2	2	2	2

(1– Low 2 –Moderate 3 – High)

- **PO/ PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO10**	**PSO1**
	3	2	2	2	2
Justification for correlation	The basic knowledge of data structures and algorithms is essential to understand the fundamentals of B+ Tree indexing in database systems.	The student should be able to analyze the indexing needs and choose or construct appropriate B+ Tree structures for efficient query processing.	The student will apply B+ Tree indexing principles to design and implement solutions for real-time data access problems, following standard engineering practices.	The student will be able to effectively document, present, and communicate technical details related to B+ Tree indexing through reports or collaborative projects.	The student will be able to solve the contemporary problems by applying the knowledge of database indexing and optimization in database management systems using B+ Trees.

- **Images/Screenshot of the practice:**

Figure 1. Students to make the group on own

Figure2. Problem statements were solved by the students

- **Reflective Critique:**

  ✓ ***Feedback of practice from students and other stakeholders:***
    - The students felt that the self-learning skills improved due to this type of activity.
    - These types of activities are interesting which involve us to learn and teach the concepts to our group members.
    - It helps them to improve the communication skills i.e. interacting effectively with the group and summarizing the concept.

  ✓ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

  Through this activity the students are made to work in a team and share their ideas to others and also improve their communication skills. Thus the students were made to understand that the team work will give a better solution.

- ***Challenges faced in implementation:***
  - **Time management:** while discussing the concepts and teaching to the other group members, the time allotted was not enough.

- **References:**

  - https://www.ritrjpm.ac.in/images/computer-science/
  - https://www.valamis.com/hub/collaborative-learning
  - https://teaching.cornell.edu/teaching-resources/engaging-students/collaborativelearning
  - https://www.evergreen.edu/sites/default/files/facultydevelopment/docs/WhatisColl aborativeLearning.pdf